

DAY 02- COMPUTER NETWORKING ASSIGNMENT

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Course:- BCA Specialisation in cybersecurity

Day:- 02

1. What is a Computer Network?

A Computer Network is a digital telecommunication network which allows nodes to share resources.

2. What is the difference between a client and a server?

Client	Server
• Requests a service/resource • Usually used by the end user • Ex:- Web Browser	• Provides the services/resources • Usually provides services to multiples clients • Ex:- Web server

Ex:- When you open Google in Chrome, Chrome acts as the clients and Google's server provides the requested webpage.

3. What is a LAN?

LAN (Local Area Network) is a network that covers a small geographical area such as a home, school, office, or computer lab.

Ex:- Computers connected within a college computer lab.

4. What is a WAN?

WAN (Wide Area Network) is a network that covers a large geographical area such as cities, countries, or even the whole world.

Ex:- The Internet is the largest example of a WAN.

5. What is the purpose of a switch?

A switch connects multiple devices within a LAN and forwards data to the specific device for which it is intended, using MAC addresses.

Ex:- Connecting 20 computers in an office network.

6. What is the purpose of a router?

A router connects different networks and forwards packets between them using IP addresses.

Ex:- Your home router connects your home LAN to the Internet.

7. What is a firewall?

A firewall is a security mechanism that monitors and controls incoming and outgoing network traffic based on predefined rules.

It helps to prevent unauthorised network access.

8. What is an IP address?

An IP (Internet Protocol) address is a logical address assigned to a device on a network so that the device can be identified and communicate with other devices.

Ex. IPv4:- 192.168.1.10

9. What is a MAC address?

A MAC (Media Access Control) address is a unique hardware/network-interface identifier used for communication within a local network.

Ex.:- 00-1A-2B-3C-4D-5E

10. What is a default gateway?

A default gateway is the device, usually a router, that forwards traffic from your local network to other networks.

11. What is DNS?

DNS (Domain Name System) converts human-readable domain names into IP addresses.

Ex.:- google.com — DNS — IP address

Without DNS, users would have to remember IP addresses instead of domain names.

12. Why is networking knowledge important for cybersecurity?

Networking knowledge is important for cybersecurity because most cyberattacks involve networks or network-connected systems.

It helps a cybersecurity professional understand:

- IP addresses
- Ports and Protocols
- Network traffic
- Firewalls
- DNS
- Routers and switch
- Packet analysis
- Network attack
- Intrusion detection
- Network monitoring

13. Write your computer's private IPv4 address.

IPv4:- 10.83.207.97

14. Write your default gateway.

gateway:- 10.83.207.244

15. Write your DNS server.

nameserver[0] : 2409:40d4:34:baa4::da
nameserver[1] : 10.83.207.244

```
[rishabhraj@Rishabhs-MacBook-Air ~ % ipconfig getifaddr en0
10.83.207.97
[rishabhraj@Rishabhs-MacBook-Air ~ % route -n get default | grep gateway
gateway: 10.83.207.244
[rishabhraj@Rishabhs-MacBook-Air ~ % scutil --dns | grep 'nameserver\[0-9\]*\]'
nameserver[0] : 2409:40d4:34:b4a4::da
nameserver[1] : 10.83.207.244
nameserver[0] : 2409:40d4:34:b4a4::da
nameserver[1] : 10.83.207.244
rishabhraj@Rishabhs-MacBook-Air ~ %
```

Use Ping Command:-

```
PING google.com (142.250.29.139): 56 data bytes
64 bytes from 142.250.29.139: icmp_seq=0 ttl=106 time=34.039 ms
Request timeout for icmp_seq 1
64 bytes from 142.250.29.139: icmp_seq=2 ttl=106 time=35.091 ms
64 bytes from 142.250.29.139: icmp_seq=3 ttl=106 time=38.420 ms
Request timeout for icmp_seq 4
Request timeout for icmp_seq 5
64 bytes from 142.250.29.139: icmp_seq=6 ttl=106 time=27.123 ms
64 bytes from 142.250.29.139: icmp_seq=7 ttl=106 time=34.207 ms
Request timeout for icmp_seq 8
64 bytes from 142.250.29.139: icmp_seq=9 ttl=106 time=39.296 ms
64 bytes from 142.250.29.139: icmp_seq=10 ttl=106 time=34.813 ms
64 bytes from 142.250.29.139: icmp_seq=11 ttl=106 time=39.214 ms
Request timeout for icmp_seq 12
64 bytes from 142.250.29.139: icmp_seq=13 ttl=106 time=37.524 ms
64 bytes from 142.250.29.139: icmp_seq=14 ttl=106 time=58.741 ms
Request timeout for icmp_seq 15
64 bytes from 142.250.29.139: icmp_seq=16 ttl=106 time=36.681 ms
64 bytes from 142.250.29.139: icmp_seq=17 ttl=106 time=42.730 ms
Request timeout for icmp_seq 18
Request timeout for icmp_seq 19
64 bytes from 142.250.29.139: icmp_seq=20 ttl=106 time=38.587 ms
64 bytes from 142.250.29.139: icmp_seq=21 ttl=106 time=33.163 ms
64 bytes from 142.250.29.139: icmp_seq=22 ttl=106 time=27.488 ms
```

Use tracert Command:-

```
C:\Users\kk861>tracert google.com
```

```
Tracing route to google.com [2404:6800:4013:813::8b]  
over a maximum of 30 hops:
```

1	3 ms	2 ms	3 ms	2409:4085:2d15:591d::7a
2	*	*	*	Request timed out.
3	43 ms	67 ms	69 ms	2405:200:33f:eeee:20::1092
4	38 ms	69 ms	51 ms	2405:200:801:4600::42e
5	*	*	*	Request timed out.
6	*	*	*	Request timed out.
7	185 ms	79 ms	*	2405:200:802:760::8
8	84 ms	*	89 ms	2405:200:802:760::8
9	*	*	*	Request timed out.
10	100 ms	75 ms	57 ms	2404:6800:8285:280::1
11	82 ms	70 ms	96 ms	2404:6800:8285:280::1
12	78 ms	74 ms	58 ms	2001:4860:0:1::6d0
13	77 ms	82 ms	81 ms	2001:4860:0:1::a7ec
14	112 ms	76 ms	78 ms	2001:4860::c:4004:53bf
15	73 ms	79 ms	71 ms	2001:4860::c:4004:2137
16	133 ms	144 ms	74 ms	2001:4860::cc:4004:1f81
17	107 ms	101 ms	81 ms	2001:4860:0:1::97b5
18	*	*	*	Request timed out.
19	*	*	*	Request timed out.
20	*	*	*	Request timed out.
21	*	*	*	Request timed out.
22	*	*	*	Request timed out.
23	*	*	*	Request timed out.
24	*	*	*	Request timed out.
25	*	*	*	Request timed out.
26	*	*	*	Request timed out.
27	84 ms	86 ms	117 ms	lcdels-in-f139.1e100.net [2404:6800:4013:813::8b]

```
Trace complete.
```